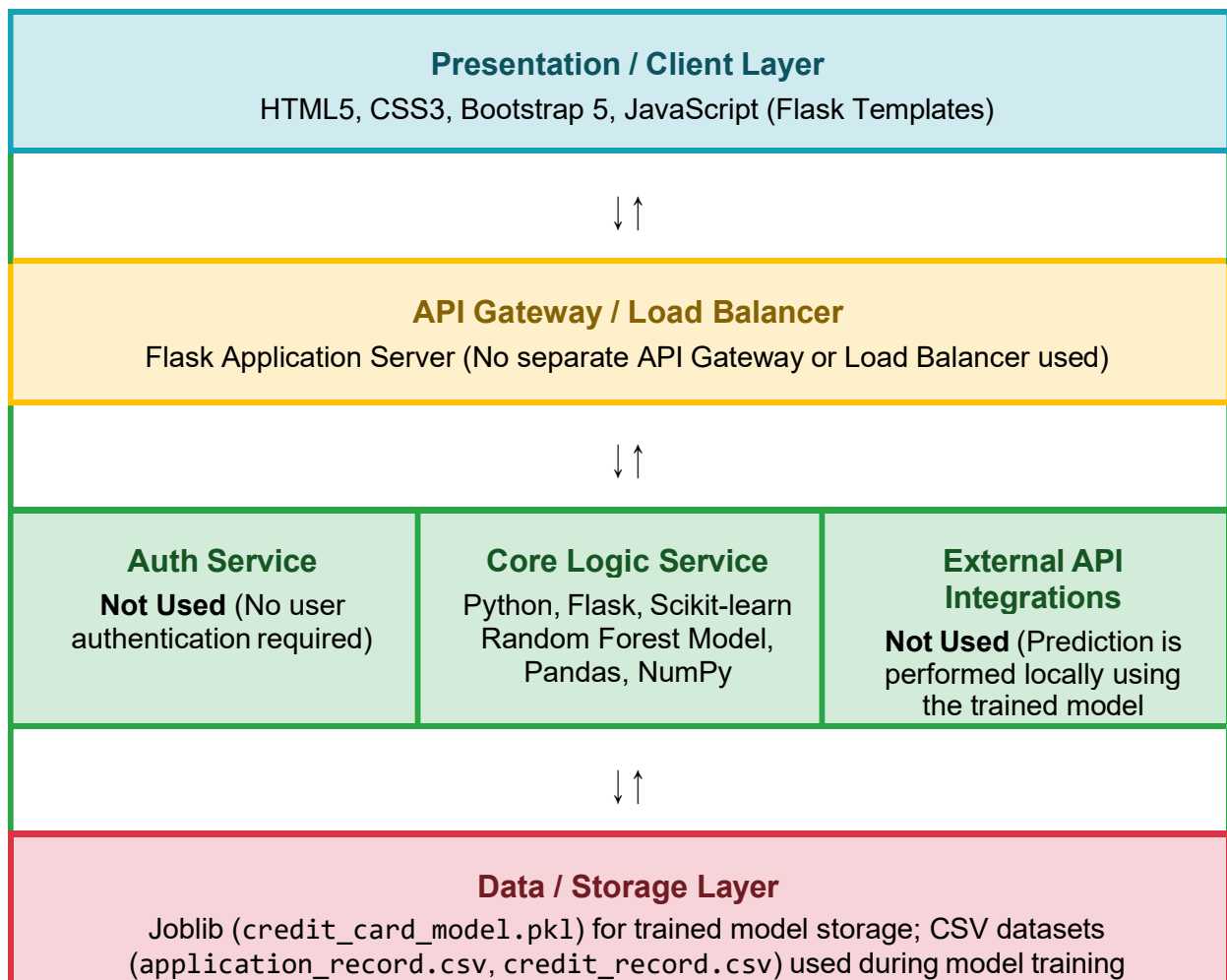


Solution Architecture

Date	15 July 2026
Team ID	SWTID-2026-8239
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface for applicants to enter personal, financial, and employment details and displays the credit card approval prediction result.	HTML5, CSS3, Bootstrap 5, JavaScript, Flask Templates
API Gateway	Acts as the entry point for handling HTTP requests from the user interface and routes them to the Flask backend for processing.	Flask Web Server
Microservices	Performs data preprocessing, loads the trained Random Forest model, processes applicant information, and generates the approval prediction.	Python, Flask, Scikit-learn, Pandas, NumPy, Joblib
Database / Storage	Stores the trained machine learning model used for prediction. The datasets are used only during model training and are not accessed during prediction.	Joblib (credit_card_model.pkl), CSV files (application_record.csv, credit_record.csv)